# Calvin Li

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## University of California, Davis

2014 - B.S. Computer Science
2016 - M.S. Computer Science

**Research Project: Random testing of database engines**
June - September 2014

- Randomly generate test cases (SQL queries) in order to look for bugs in database engines
- Test for correctness by transforming queries to equivalent ones and comparing
- Can also test correctness by running queries on multiple engines

## Programming Languages and Technologies

**Professional Experience**

| | |
|---|---|
| Python | DataDog |
| C# | Google Cloud |
| Azure | GitLab |
| Kubernetes | PagerDuty |

**Proficient**

| | |
|---|---|
| Java | LaTex |
| Kotlin | |

**Experience**

| | |
|---|---|
| C | JavaScript |
| Swift | Helm |
| C++ | |

## Personal Projects

**Quest**
github.com/calvin-li/Quest

- I do Android development as a hobby
- Entirely self-taught
- This is an app I made to help me track tasks using Android's notification system
- Source code on Github, also on the Play Store

## Zonar Systems, Inc

March 2020 - March 2023
**Software Engineer II**, OEM team

- Maintained and improved existing OEM services
  - Services generally ingested data from partner hardware, processed, then inserted data into Zonar's systems
  - Required collaborating directly with multiple different partner companies (eg DTNA, Cummins), each with their own hardware and systems
  - Learned the systems well enough to be part of the on-call rotation within 3 months of joining
  - Improved DevOps and helped uplift the services to the cloud
- Helped develop Light Duty TCU (Telematics Control Unit) system
  - A simpler, cheaper version of Zonar's existing product geared towards light vehicles (eg cars and pickups) instead of large trucks
  - Collaboration with parent company (Continental Ag), who developed the hardware
  - System developed from the ground up as a part of a larger OEM platform, designed to eventually replace all the various services we currently have
  - Saw the product through from planning to launch (Dec. 2021) and beyond
  - Developed a load testing system that takes previously recorded input and replays it to simulate a system under load
- Helped plan and develop the OEM Platform
  - This is a restructure of our services to be more generalized and use more shared code
  - Beginning with Light Duty TCU, and then integrating out older services
  - The idea is to create a common framework to ingest data from various sources, process it, and then send it to our backend
- OEM systems made up of micro-services, mix of Google Cloud Functions and containers in GKE. Data handled using MemoryStore (Google Redis) and Google DataStore.
  - Python used throughout
  - Heavy use of GitLab CI for continuous pre-prod deployments, and quick prod releases
  - Use Terraform to manage Google Cloud Infrastructure through code
- They're based in Seattle, but have been primarily remote since I joined

## Zume, Inc

November 2019 - March 2020
**Software Engineer**

- Worked on web app for tracking production at packaging factory
  - Improved CI/CD of system, bringing it in line with company wide standards
  - Learned some web development and containers

## Microsoft Corporation

August 2017 - June 2019
**Software Engineer**, Microsoft Graph

- Proxy API for AAD Graph and other Microsoft APIs
- Split time between feature development and maintenance of 4 nines uptime
- Expanded service into new cloud environment (US government)
- Took ownership of our integration testing service
- Part of the on-call rotation
  - Responded to incidents from automated monitoring or users 24/7
  - Used experience to analyze health of service and make improvements to both project setup and livesite setup
  - Designed and implement improvements to our alerting and troubleshooting.

June 2016 - August 2017
**Software Engineer**, Azure AD Graph

- Rest API for Azure Active Directory
- Added publisher domain feature to Applications API
  - Lets application owners specify their a domain to show users when asking for permissions
  - Responsible for implementation and test design
- Maintained the product's C# library
  - Helps developer make requests to the API without having to craft HTTP Rest requests
  - Inherited sole ownership upon joining team
  - Fixed bugs, cleaned up the code, and added automation
  - Worked with users to transition them to successor product (Microsoft Graph)

June 2015 - September 2015
**Software Engineer Intern**, Microsoft Authenticator App

- iOS app for Microsoft's MFA service
- Added usability features (fingerprint authentication and Apple Watch integration)

- Main app coded in Objective-C, watch app coded in Swift